

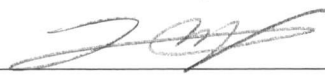
UNIVERSITY OF TORONTO
Faculty of Arts and Science
MAT133Y1Y: Calculus and Linear Algebra for Commerce
Term Test 1

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Duration: 105 minutes

Aids Allowed: Approved Scientific Calculator

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Instructions

- **Keep this booklet closed**, until an invigilator announces that the test has begun. However, you may fill out your information in the box above at any time.
- Please place your **student ID card** on your desk, so that is easy for an invigilator to check without disturbing you during the test.
- **Turn your cell phone completely off** (not just in vibrate mode) and store it with your other belongings under your desk or on the back of your chair.
- For all multiple choice questions, fill in your answer fully using dark pencil or ink (unfilled ☐ ; filled ☒).
- Questions for which you may select multiple correct answers use squares (unfilled ☐ ; filled ☒).
- Please write your answers into the boxes. Ample space is provided within each box, however, if you must use additional space, please use the blank page at the end of this booklet, and clearly indicate in the given box that your answer is **continued on the last page**.
- This test booklet contains 12 pages and 11 questions. If your booklet is missing a page, please raise your hand to notify an invigilator as soon as possible.
- **Do not remove any page from this booklet.**
- Scrap paper will be distributed separately. Please raise your hand to ask an invigilator if you need more scrap paper. **Anything you write on the scrap paper will not be marked.**

The sum of a finite geometric series is given by

$$\sum_{i=0}^{n-1} aR^i = \frac{a(1 - R^n)}{1 - R},$$

where a and R are real numbers, with $R \neq 1$.

Page:	2	3	4	5	7	8	9	Total
Points:	9	8	8	12	8	9	6	60
Score:								

1. (6 points) Toronto Public Health (TPH) measures E. coli levels daily at public beaches to assess safety for public swimming. Below are measurements taken at Sunnyside Beach at time t days after a flood.

Time t (days)	0	1	2	3
E. coli level L (E. coli per 100 mL)	1020	415	110	90

Consider the E. coli level L as a function of time t , $L = f(t)$.

- (a) Find the average rate of change of $f(t)$ between $t = 0$ and $t = 3$. Interpret the result for the public.

$$\frac{1020 - 90}{0 - 3} = \frac{930}{-3} = -310 \text{ (E. coli per 100 mL / day)}$$

change in E. coli level
change in days

During the 3 days after the flood, the E. coli levels at Sunnyside Beach were

- ☐ increasing
☒ decreasing

at an average rate of

310

- ☐ E. coli
☐ E. coli per 100 mL
☒ E. coli per 100 mL per day

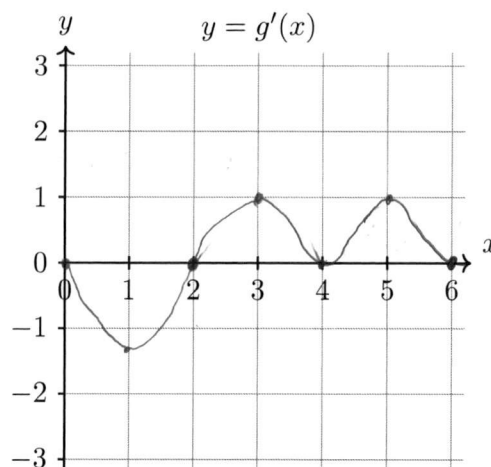
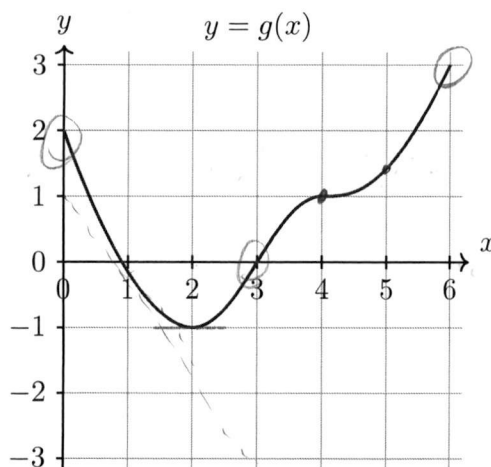
- (b) Which of the following statements about f' and f'' are true, based on the given data?

- ☐ Since $f'(t)$ is increasing, $f''(t)$ must be positive. ✗ ☐ Since $f'(t)$ is positive, $f''(t)$ must be increasing. ✗
☐ Since $f'(t)$ is decreasing, $f''(t)$ must be negative. ✗ ☐ Since $f'(t)$ is negative, $f''(t)$ must be decreasing. ✗
☒ None of the above.

- (c) Which of the following sentences best expresses the result from part (b) for the general public?

- ☐ The E. coli levels are dropping faster and faster each day.
☒ The E. coli levels are dropping slower and slower each day. ✓
☐ The E. coli levels are rising faster and faster each day.
☐ The E. coli levels are rising slower and slower each day.
☐ None of the above.

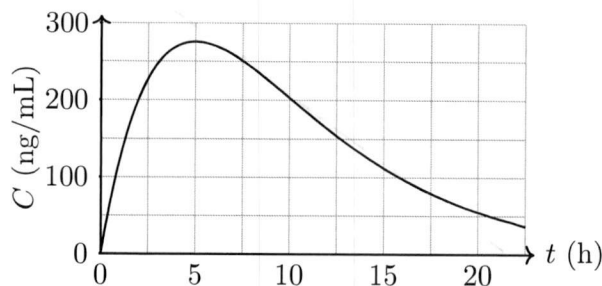
2. (3 points) A function g is graphed on the left. Sketch the graph of its derivative, g' , on the right.



$$-\frac{4}{3} = -1.332$$

3. (8 points) A patient takes *Achooziprine*, a new oral allergy medicine.

The concentration of the *Achooziprine* in the patient's bloodstream C (in ng/mL) is a function of time t (in hours) since taking the drug. The graph of this function, $C = f(t)$, is shown on the right.



- (a) Estimate $f'(7.5)$. Choose the option below that is closest to your estimate.

- | | | | | |
|--|---|--|--|---|
| ☐ -17 ng/mL | ☐ -10 ng/mL | ☐ -1 ng/mL | ☐ 33 ng/mL | ☐ 250 ng/mL |
| ☐ -17 ng/h | ☐ -10 ng/h | ☐ -1 ng/h | ☐ 33 ng/h | ☐ 250 ng/h |
| ☒ -17 $\frac{\text{ng/mL}}{\text{h}}$ | ☐ -10 $\frac{\text{ng/mL}}{\text{h}}$ | ☐ -1 $\frac{\text{ng/mL}}{\text{h}}$ | ☐ 33 $\frac{\text{ng/mL}}{\text{h}}$ | ☐ 250 $\frac{\text{ng/mL}}{\text{h}}$ |

- (b) At approximately which value of t does the graph have an inflection point? What is the significance of the inflection point in this context?

The graph has a point of inflection at time $t \approx$ ☐ 5 h ☒ 10 h ☐ 15 h ☐ 20 h.

At this time, the rate of change of the *Achooziprine* concentration (i.e. $\frac{dC}{dt}$) is

- ☐ greatest ☒ zero ☐ least

- (c) From further data, it is known that $f(28) = 15$, $f'(28) = -3$, and for $t > 25$, $f''(t) > 0$. Based on this, which of the following are possible concentrations (in ng/mL) of *Achooziprine* in the patient's bloodstream 30 hours after taking the drug? Colour in the squares for **all possible** values and explain your reasoning clearly for another MAT133 student.

- ☒ 3 ☒ 4 ☒ 5 ☒ 6 ☒ 7 ☒ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12

Explanation:

$t > 25$ and $f''(t) > 0$ means that the rate of change of concentrations is increasing. $f'(28) = -3$ means that one extra hour, the concentration will drop by 3, and based on $f(28) = 15$, $f(29) = 12$, since the rate of change increase, $f'(29) < -3$, so after 30 hours, the concentration should be at least lower than 9. Therefore any value below 9 is possible.

- (d) In addition to the information provided in (c), suppose you also know that $f'(30) = -1.5$.

True or False: In this case, we can determine the value of $f(30)$ exactly.

- ☐ True, and $f(30) = 12$. ☐ True, but $f(30) \neq 12$. ☒ False, we cannot determine $f(30)$ exactly.

4. (5 points) Poppy's Popcorn sells gourmet bags of popcorn at movie theatres. The number of bags they sell each day, q , depends on the price, p (in dollars per bag). Their daily revenue R (in dollars) also depends on the price p . The owner, Poppy, is unfamiliar with calculus.

(a) Interpret the meaning of $\left. \frac{dR}{dp} \right|_{p=13} = -400$ for Poppy. Fill in the box for **all** correct interpretations.

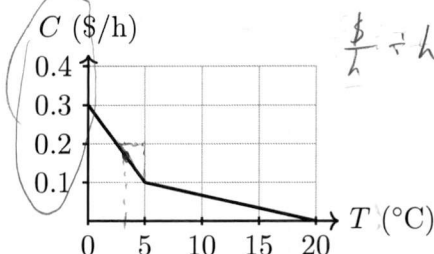
- ☒ If Poppy charges \$13 per bag, her marginal revenue is $-\$400$ per bag.
- ☐ If Poppy charges \$13 per bag, her marginal revenue is \$400 per bag.
- ☐ If Poppy increases the price from \$13 to \$14 per bag, her revenue will decrease by about \$400.
- ☐ If Poppy increases the price from \$13 to \$14 per bag, her revenue will increase by about \$400.

(b) From consumer data, it's discovered that when the price of Poppy's popcorn is 8 dollars per bag, Poppy sells 500 bags daily and $\left. \frac{dq}{dp} \right|_{p=8} = -20$. Based on this, calculate the value of $\left. \frac{dR}{dp} \right|_{p=8}$. Explain your reasoning and then write your final answer in the box below. Do not include units.

$\left. \frac{dq}{dp} \right|_{p=8} = -20$ means when p is 8, one more increase in price will cause 20 reduction on the number of bags sales. $\left. \frac{dR}{dp} \right|_{p=8}$ is change in Revenue with respect to p . If p increase by 1 dollar Revenue will change by $-20 \cdot 9 = -180$ dollar. If price increase by 1 dollar to 9 dollar, q will lose 20 and Revenue will change by $p \cdot q$ which is -180 .

$\left. \frac{dR}{dp} \right|_{p=8} = -180 \text{ dollar}$

5. (3 points) The hourly heating cost C (in \$/h) for a building is a function of the outside temperature T (in $^{\circ}\text{C}$), graphed below. The outside temperature T varies as a function of time t in hours since midnight.



$$\frac{\$}{h} \div h = \frac{\$}{h^2}$$

t (h)	6	8	10	12
T ($^{\circ}\text{C}$)	2.2	3.4	8.1	11.3
$\frac{dT}{dt}$ ($^{\circ}\text{C}/\text{h}$)	0.8	2.5	3.2	1.6

At what rate were the hourly heating costs C changing with respect to time at 8 AM?

$T = f(t)$ $C = g(T) \Rightarrow C = g(f(t))$ so the rate of change of hourly heating cost C respect to time at 8 AM is C' when $t=8$

$C = g(f(t))$ $f'(8)$ is $\left. \frac{dT}{dt} \right|_{t=8}$ which is 2.5 $t=8$ T is 3.4 as given

$C' = g'(f(t)) \cdot f'(t)$ $g'(3.4) = \frac{-0.1}{2.5} = -0.04$

$C' = g'(3.4) \cdot 2.5 = -0.04 \cdot 2.5$

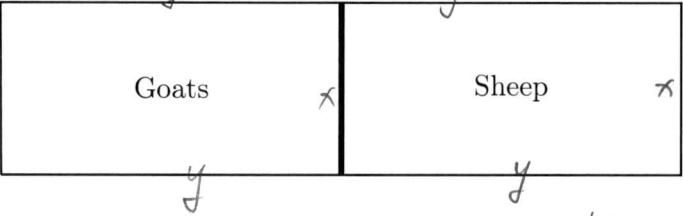
$\left. \frac{dC}{dt} \right|_{t=8} \approx$

-0.1	$\$/h^2$
(number)	(units)

6. (12 points) A local farmer has asked your consulting firm to design a rectangular pen for their goats and sheep. The farmer needs the pen to have two equally-sized sections, one for goats and one for sheep, as shown in the figure below. The outer walls of the pen may be built using materials that cost \$20 per metre. The dividing wall (marked with a thicker line in the figure) must be constructed with materials that cost \$60 per metre. You have a total budget of \$50,000 to spend on the materials. Use calculus to find the largest possible total area enclosed by the fences.

1. Define your variables clearly on the picture provided.
2. If you define a function, indicate its domain.
3. Use full sentences to motivate and explain your process.
4. Explain why your answer is the maximum.

If you need more space for your solution, you may also use the next page.



total budget

$$A(x) = 1250x - \frac{5}{2}x^2$$

$$A'(x) = 1250 - 5x$$

$$0 = 1250 - 5x$$

$$x = 250 \text{ meter}$$

sub into $y = 625 - \frac{5}{4}x$ for y value

$$y = 625 - \frac{5}{4} \cdot 250$$

$$y = 312.5 \text{ meter}$$

Area = $x \cdot y$

$$= 250 \cdot 312.5$$

$$= 78125 \text{ meter square}$$

checking:

$$20(4 \times 312.5 + 2 \cdot 250) + 60 \cdot 250 = 50000$$

let the width of each section be x
let the length of each section be y
Cost function will be

$$C = 20(4y + 2x) + 60x$$

$$50000 = 20(4y + 2x) + 60x$$

$$50000 = 80y + 40x + 60x \quad \text{Divid 20}$$

$$2500 = 4y + 2x + 3x$$

$$2500 = 4y + 5x$$

$$4y = 2500 - 5x$$

$$y = 625 - \frac{5}{4}x \rightarrow \text{relation of } x \text{ and } y$$

let Area function be $x \cdot 2y$

$$A(x) = x \cdot 2(625 - \frac{5}{4}x)$$

$$A(x) = x \cdot (1250 - \frac{5}{2}x)$$

$$A(x) = 1250x - \frac{5}{2}x^2$$

In order to find maximum area, when the first derivative of the function is zero or undefined, let

$$A'(x) = 0$$

Answer: The largest possible total area enclosed by the fences is 78125 m².

If needed, round to the nearest integer.

If you need extra space for your solution, you may use the next page.

This is extra space for your work on Problem 6, if you need it.

$$y = 625 - \frac{5}{4}x$$

$$y > 0$$

$$625 - \frac{5}{4}x > 0$$

$$-\frac{5}{4}x > -625$$

$$x < (625) \cdot \left(-\frac{4}{5}\right)$$

$$x < 500$$

$$0 < x < 500$$

$$2500 = 4y + 5x$$

$$5x = 2500 - 4y$$

$$x = 500 - \frac{4}{5}y$$

$$x > 0$$

$$500 - \frac{4}{5}y > 0$$

$$-\frac{4}{5}y > -500$$

$$y < (-500) \cdot \left(-\frac{5}{4}\right)$$

$$y < 625$$

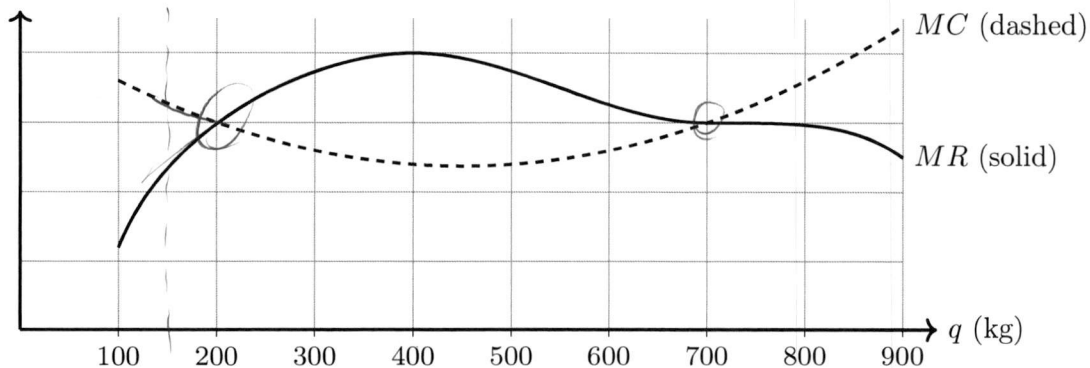
$$0 < y < 625$$

$$A'(x) = 1250 - 5x$$

$$A''(x) = -5$$

$$A''(x) < 0 \text{ max}$$

7. (8 points) Billy Bonka produces and sells gummy bears in bulk, by the kilogram. The marginal cost and marginal revenue for producing and selling q kilograms of gummy bears in a month are graphed below, where $100 \leq q \leq 900$.



- (a) On $100 \leq q \leq 900$, at what approximately which value(s) of q is the profit **locally maximized**? There may be multiple local maxima; please fill in one box for each. Then explain your answer in the space below, for an audience of fellow MAT133 students.

☐ 100 ☒ 200 ☐ 300 ☐ 400 ☐ 500 ☒ 700 ☐ 900

Explanation:

In order to find at which value of q is the profit locally maximized we need to find when marginal profit is zero. Marginal profit is zero means $MR - MC$ is zero which $MR = MC$. $MR = MC$ when $q = 200$ and $q = 700$. At $q = 200$ profit reach locally max for the first and at $q = 700$ profit reach locally max for the last time as graph shown.

- (b) Suppose that Billy Bonka is currently producing and selling 150 kg of gummy bears per month. If he increases his production and sales to 150.1 kg, what will happen to his profit?

- ☒ His profit will increase.
☐ His profit will stay about the same.
☐ His profit will decrease.
☐ More information is needed to answer.

8. (2 points) A client would like to invest into a savings plan. Among the following interest schedules, which would you recommend?

- ☐ The account will receive 2% interest on its balance at the end of each month.
☐ The account will receive 4% interest on its balance every two months.
☐ The account will receive 8% interest on its balance every four months.
☒ The account will receive 26% interest on its balance at the end of each year.
☐ All of the above are equally good savings plans.

9. (7 points) Maryam is going to receive a scholarship of \$5000 per year, to be received at the beginning of each year of her undergraduate studies. Suppose that she invests all of this scholarship money into a bank account that collects interest at an annual rate of 4%, compounded continuously. No other deposits or withdrawals are made to this account. How much money will be in her account right before she receives the fourth scholarship? Right after? **Organize your work and explain your reasoning clearly** for an audience of fellow MAT133 students. Please round your answers to the nearest \$0.01.

First: 5000 first deposit

Second: $5000(1.04) + 5000$ second deposit + before $\times$ interest rate

Third: $5000(1.04)^2 + 5000(1.024) + 5000$ third deposit + before $\times$ interest rate

Fourth: $5000(1.04)^3 + 5000(1.024)^2 + 5000(1.024) + 5000 = 21232.32$

fourth deposit + before $\times$ interest rate.

right before the fourth deposit will be the amount in account after fourth year subtract the fourth deposit.

$$21232.32 - 5000 = 16232.32 \text{ dollar}$$

Right before is the previous amount time interest rate

Right after is Right before plus a new deposit.

Right before is Right after without the new deposit.

Right **before** she receives the fourth scholarship, her account balance will be \$

16232	32
(dollars)	(cents)

Right **after** she receives the fourth scholarship, her account balance will be \$

21232	32
(dollars)	(cents)

10. (5 points) A local company, BoxCo, wants to fund scholarships for future U of T students. Starting today, they will give an \$8000 scholarship to a U of T student every year. To finance these scholarships forever, BoxCo will deposit a lump sum of money today into a bank account with an annual interest rate of 3%, compounded continuously.

(a) What is the present value of all scholarships (assuming they'll be given ~~every two years~~ forever)?

- | | |
|--|---|
| ☐ $8000 + 8000(0.03)^1 + 8000(0.03)^2 + \dots$ | ☐ $8000 + 8000(0.03)^{-1} + 8000(0.03)^{-2} + \dots$ |
| ☐ $8000 + 8000(1.03)^1 + 8000(1.03)^2 + \dots$ | ☒ $8000 + 8000(1.03)^{-1} + 8000(1.03)^{-2} + \dots$ |
| ☐ $8000 + 8000e^{0.03(1)} + 8000e^{0.03(2)} + \dots$ | ☐ $8000 + 8000e^{-0.03(1)} + 8000e^{-0.03(2)} + \dots$ |
| ☐ $8000 + 8000e^{1.03(1)} + 8000e^{1.03(2)} + \dots$ | ☐ $8000 + 8000e^{-1.03(1)} + 8000e^{-1.03(2)} + \dots$ |

(b) How much money does BoxCo need to deposit as a lump sum in their account today, to ensure that they can keep funding these scholarships forever? Please show your reasoning and round your answer to the nearest \$0.01.

1. $x = 8000$

2. $(x - 8000)(1.03) = 8000$

3. $(x - 8000)(1.03)^2 = 8000(1.03)$

Based on the numbers

$$\frac{8000}{1 - (1.03)^{-1}} = 274666.67 \text{ dollar}$$

Since forever using infinite series

BoxCo needs to deposit at least \$ 274666 . 67 . in their account today.

(dollars)

(cents)

11. (1 point) Please take a moment to reflect. There are no right or wrong answers to this question, however, you will receive a point for answering fully.

(a) Which question did you feel the most confident about solving?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☒ 7 ☐ 8 ☐ 9 ☐ 10

(b) Which question did you find the most challenging?

- ☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

(c) Before you leave the test room, please check all of your work. Find at least one solution that you can make an improvement on. Briefly summarize the improvement you made below.

I made an improvement in my solution to Question

7

. My improvement was... *more detailed explanations.*

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This space is provided for additional space for your solutions if needed. If you must use this space, be sure to clearly indicate in the original question that your solution is continued on Page 11.

Do not remove this page from the test booklet.

This space is provided for additional space for your solutions if needed. If you must use this space, be sure to clearly indicate in the original question that your solution is continued on Page 12.

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